

Name: _____

Date: _____

DIRECTIONS

This assessment covers lessons 4.09.01 to 4.09.04 (entire unit).

Show your work in the box. Write your final answer on the line.

Read every problem carefully. Some problems have more than one part — answer ALL parts.

Take your time. Check your work before you turn in.

1 Mode, Median, Range — small set

Data: 4, 3, 5, 4, 3, 2

(a) Mode (b) Median (c) Range

ORDER FIRST

(a) ____ (b) ____ (c) ____

2 Mode, Median, Range — unsorted set

Data: 2, 3, 9, 2, 5, 14, 2, 8, 9, 0, 2, 9

(a) Reorder the data.

(b) Mode (c) Median (d) Range

REORDER FIRST

(a) ____ (b) ____ (c) ____ (d) ____

3 Line plot — build a frequency table

Old Faithful durations: 2, 3, 4, 4, 2, 3, 5, 4, 3, 3, 4, 5, 2, 4, 3, 5, 4, 3, 4, 2

Build a frequency table for 2, 3, 4, 5 (count each value).

TALLY EACH VALUE

Counts: 2: ____ 3: ____ 4: ____ 5: ____

4 Line plot — read the plot

A line plot of travel times shows: 10 (3), 15 (5), 20 (5), 25 (6), 30 (4), 35 (2), 40 (2), 45 (1), 50 (1).

(a) How many students total?

(b) Mode (c) Range

(d) What fraction of students took MORE THAN 30 minutes? (simplest form)

SUM ALL X COUNTS; PART/TOTAL

(a) ____ (b) ____ (c) ____ (d) ____

5 Line plot — fractional values

Twelve students reported sleep hours (rounded to quarter hours):

9, $8\frac{3}{4}$, $9\frac{1}{4}$, 8, 9, $9\frac{1}{2}$, $9\frac{1}{4}$, $8\frac{3}{4}$, 9, 10, $8\frac{1}{2}$, $9\frac{1}{2}$

(a) How many students did NOT get exactly 9 hours?

(b) What fraction got AT LEAST 9 hours? (simplest form)

COUNT $\geq$ 9; PART/TOTAL

(a) ____ (b) ____

6 Stem-and-leaf — blue whales

Data: 2, 15, 42, 42, 46, 48, 75, 77, 88, 98

(a) Build a stem-and-leaf plot.

(b) Mode (c) Median (d) Range

STEM | LEAF

(b) ____ (c) ____ (d) ____

7 Stem-and-leaf — tortoises (3-digit)

Data: 5, 58, 66, 68, 74, 75, 76, 81, 82, 91, 102

Rule: 3-digit values go under STEM 10.

- (a) Build the stem-and-leaf plot.
(b) Mode (c) Median (d) Range

STEM | LEAF; STEM 10 FOR 102

(b) ____ (c) ____ (d) ____

8 Compare two data sets

Class A's quiz scores (out of 10): 6, 7, 7, 8, 8, 9, 9, 9, 10

Class B's quiz scores (out of 10): 5, 5, 7, 8, 8, 8, 9, 10, 10

- (a) Find the MODE for each class.
(b) Find the MEDIAN for each class.
(c) Find the RANGE for each class.
(d) Which class had scores that were more CONSISTENT (less spread out)? Explain.

COMPARE BOTH SETS

(a) A: ____ B: ____ (b) A: ____ B: ____ (c) A: ____ B: ____ (d) ____

9 Pick the right plot

You want to display the masses of 25 dogs (in kg) that range from 4 kg to 62 kg.

Would a LINE PLOT or a STEM-AND-LEAF PLOT be easier to read? Explain your choice.

WRITE YOUR REASONING

My choice: ____

10 EXPLAIN — Unit 9 reflection

Across this unit you learned MODE, MEDIAN, RANGE, LINE PLOTS, and STEM-AND-LEAF PLOTS.

Pick ONE thing you learned and describe a real situation where you might use it.

WRITE YOUR REFLECTION

My explanation: